

Social Media Sentiment Analysis at Scale



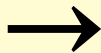
Image source: Irasutoya

Ewa Herman
Julia Krempińska
Natalia Przewdziek
Julian Sungkyun Yoo

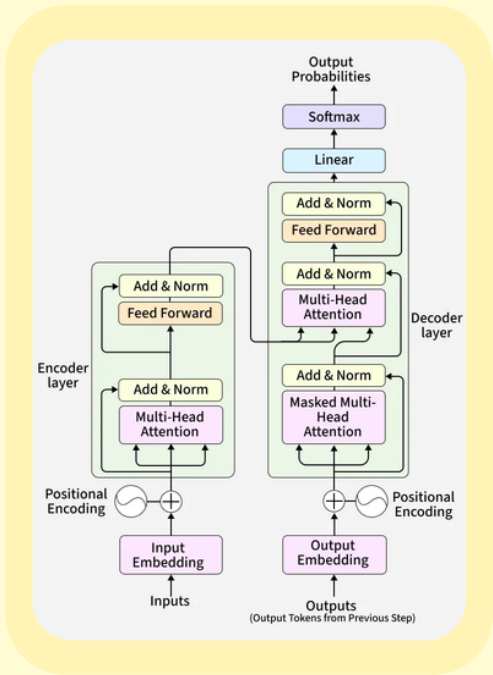
Recap

kaggle

tweet data
from cardiffnlp



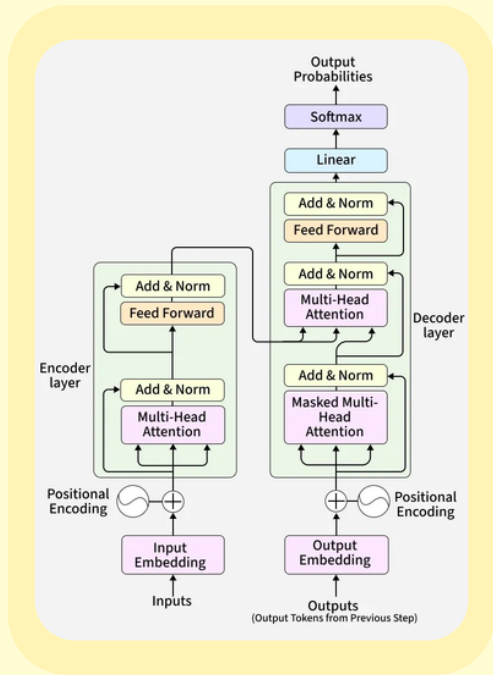
NLP + DL



LLMs

spaCy

NLP + DL



LLMs

Outline

1. Exploratory Data Analysis

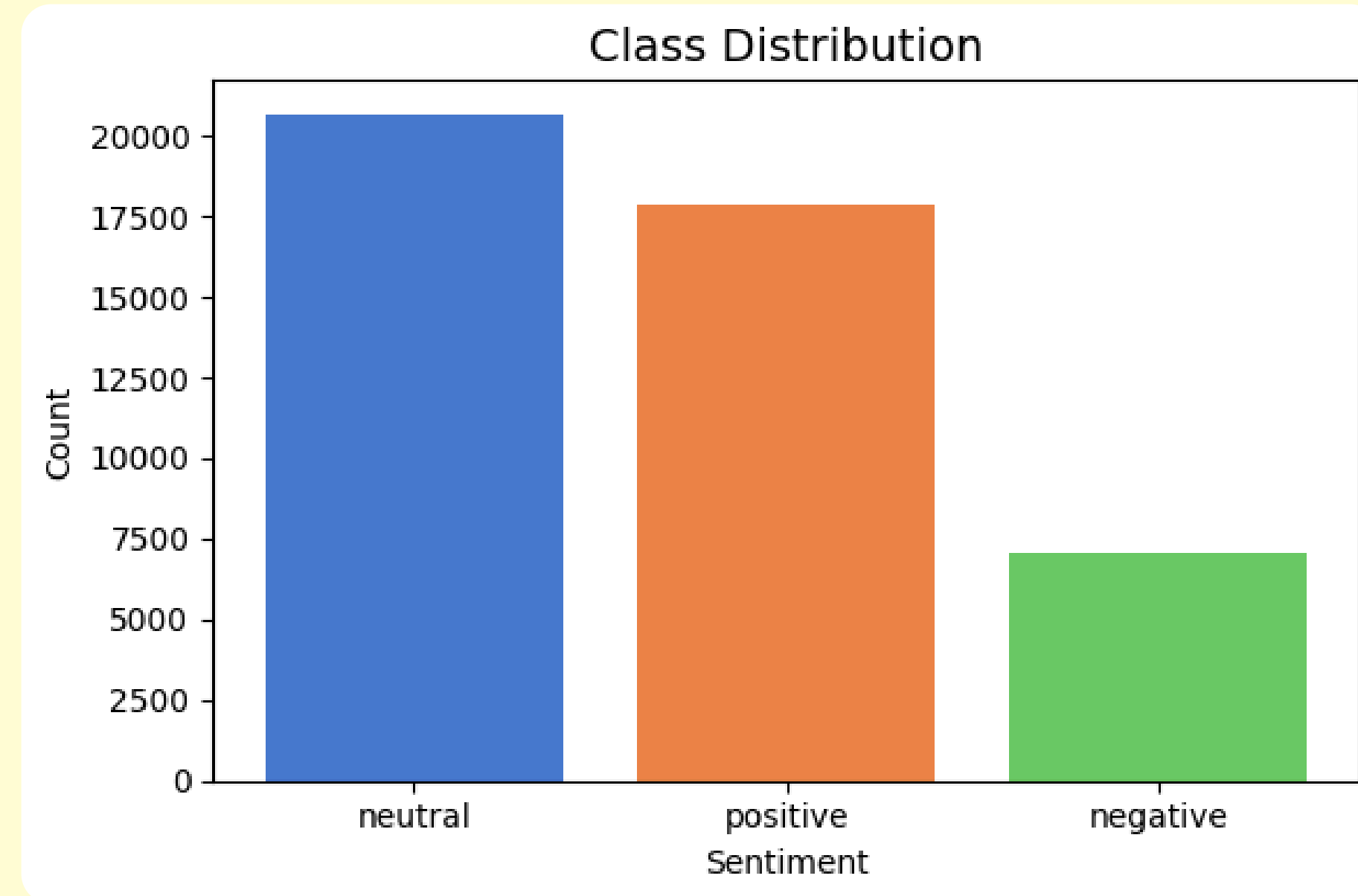
2. Traditional NLP

3. NLP + ML variants

4. LLMs

5. To be added

Exploratory Data Analysis



Dataset

- 45k tweets
- no missing data
- text + sentiment labels

Imbalance

- Neutral ~45%
- Positive ~39%
- Negative ~16%

noticeable class imbalance

Tweet Length

- ~20 words on average
- consistent across sentiments
- low variance
- no major differences between classes

NLP (Lexicon-Based Approach)

Method	Accuracy	Notes
VADER	55%	best overall
TextBlob	50%	weaker on neutral
Rule-based	51%	very basic performance

NLP (ML-Based Approach)

Method	Accuracy	Notes
Logistic Regression	62%	simple, strong baseline
SVM	62%	robust to high-dimensional data
XGBoost	59%	underperformed linear models

Analysis with Large Language Models

Models used:

1. **mBERT** (bert-base-multilingual-cased)
2. **distilbert-base-multilingual-cased**
3. **XLM-RoBERTa** (xlm-roberta-base)

Used methods:

- **Tokenisation** (SentencePiece BPE - RoBERTa & WordPiece - BERT variants)
- **Embedding** (CLS)

Intial Analysis with K-shot Inferences

zero-shot inferences

```
Zero-shot Baseline - XLM-RoBERTa
[01] False | True: neutral      | Pred: negative
      Text: @user @user what do these '1/2 naked pics' have to do with anything? They're not

[02] False | True: neutral      | Pred: negative
      Text: OH: "I had a blue penis while I was this" [playing with Google Earth VR]

[03] False | True: neutral      | Pred: negative
      Text: @user @user That's coming, but I think the victims are going to be Medicaid reci

[04] False | True: positive     | Pred: negative
      Text: I think I may be finally in with the in crowd #mannequinchallenge #grads2014 @u

[05] True  | True: negative      | Pred: negative
      Text: @user Wow,first Hugo Chavez and now Fidel Castro. Danny Glover, Michael Moore, 0

[06] False | True: neutral      | Pred: negative
      Text: Savchenko now Saakashvili took drug test live on Ukraine TV. To prove they are n
```

Zero-Shot Summary:
0.0000 accuracy

few-shot inferences

```
1-SHOT PROTOTYPE INFERENCE (Accuracy: 0.2000)
[1] False True: neutral      | Pred: positive
      Text: @user @user what do these '1/2 naked pics' have to do with anything? They're not
[2] False True: neutral      | Pred: positive
      Text: OH: "I had a blue penis while I was this" [playing with Google Earth VR]
[3] False True: neutral      | Pred: positive
      Text: @user @user That's coming, but I think the victims are going to be Medicaid reci
[4] False True: positive     | Pred: neutral
      Text: I think I may be finally in with the in crowd #mannequinchallenge #grads2014 @u
[5] False True: negative     | Pred: positive
      Text: @user Wow,first Hugo Chavez and now Fidel Castro. Danny Glover, Michael Moore, 0

3-SHOT PROTOTYPE INFERENCE (Accuracy: 0.2600)
[1] True True: neutral      | Pred: neutral
      Text: @user @user what do these '1/2 naked pics' have to do with anything? They're not
[2] False True: neutral      | Pred: negative
      Text: OH: "I had a blue penis while I was this" [playing with Google Earth VR]
[3] False True: neutral      | Pred: positive
      Text: @user @user That's coming, but I think the victims are going to be Medicaid reci
[4] False True: positive     | Pred: neutral
      Text: I think I may be finally in with the in crowd #mannequinchallenge #grads2014 @u
[5] False True: negative     | Pred: positive
      Text: @user Wow,first Hugo Chavez and now Fidel Castro. Danny Glover, Michael Moore, 0

5-SHOT PROTOTYPE INFERENCE (Accuracy: 0.2800)
[1] False True: neutral      | Pred: positive
      Text: @user @user what do these '1/2 naked pics' have to do with anything? They're not
[2] False True: neutral      | Pred: negative
      Text: OH: "I had a blue penis while I was this" [playing with Google Earth VR]
[3] False True: neutral      | Pred: negative
      Text: @user @user That's coming, but I think the victims are going to be Medicaid reci
[4] False True: positive     | Pred: neutral
      Text: I think I may be finally in with the in crowd #mannequinchallenge #grads2014 @u
[5] True True: negative      | Pred: negative
      Text: @user Wow,first Hugo Chavez and now Fidel Castro. Danny Glover, Michael Moore, 0
```

Few-Shot Summary:
1-shot: 0.2000 accuracy
3-shot: 0.2600 accuracy
5-shot: 0.2800 accuracy

Full Fine-Tuning vs PEFT LoRA Fine-Tuning

XLM-RoBERTa ~ 10 hours
on Full Fine-Tuning

Test Accuracy : 0.6628				
Test F1-Macro : 0.6650				
Classification Report:				
	precision	recall	f1-score	support
negative	0.61	0.87	0.72	3972
neutral	0.80	0.47	0.59	5937
positive	0.60	0.79	0.68	2375
accuracy			0.66	12284
macro avg	0.67	0.71	0.67	12284
weighted avg	0.70	0.66	0.65	12284

XLM-RoBERTa ~ 2.5 hours
on PEFT LoRA

Test Accuracy : 0.6252				
Test F1-Macro : 0.6244				
Classification Report:				
	precision	recall	f1-score	support
negative	0.61	0.84	0.71	3972
neutral	0.81	0.39	0.53	5937
positive	0.51	0.84	0.64	2375
accuracy			0.63	12284
macro avg	0.64	0.69	0.62	12284
weighted avg	0.69	0.63	0.61	12284

To be done

- NLP + LSTM
- Interpretability analysis
- Comparison of the methods
- Summary and notebook polishing

Thank you



Image source: Irasutoya

Reference List

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- [4] Rodriguez-Ibanez, M., et al. A review on sentiment analysis from social media platforms. (2023). Retrieved from: <https://doi.org/10.1016/j.eswa.2023.119862>
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- [6] Escalona, T. Detect sentiment from customer reviews using Amazon Comprehend. (2018). Retrieved from: <https://aws.amazon.com/blogs/machine-learning/detect-sentiment-from-customer-reviews-using-amazon-comprehend/>